Genetic Parameters and Selection Responses for Alternative Methane Trait Definitions in Pasture-Based Sheep

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Abstract

Background

Enteric methane, produced during rumen fermentation and released mainly by eructation, is a major challenge for environmentally sustainable ruminant production. Genetic selection can deliver long-term and cumulative reductions in emissions; however, methane production is positively correlated with key performance traits such as feed intake, body size and growth. Ratio- and residual-based methane definitions have therefore been proposed. The objectives of this study were to estimate genetic parameters for a range of alternative methane trait definitions in sheep, to characterise the genetic and phenotypic relationships between them and with key production traits, and to examine their use in a selection index.

Results

In total, 16,535 methane (CH_4) records from 8,354 sheep across 132 Irish flocks were analysed. Alternative methane traits were defined as ratios of methane relative to metabolic body weight (MBW), average daily gain (ADG), muscle mass and rumen volume, and as residuals of methane regressed on these traits. Heritability estimates ranged from 0.12 to 0.46. Daily methane was positively genetically correlated with MBW (rg = 0.72), live weight (0.71) and ADG (0.51). Ratio and residual definitions were negatively genetically correlated with their denominator or adjustment traits, while the residual traits remained strongly genetically correlated with absolute methane (rg = 0.78–1.00). In a selection index of CH_4, MBW and carbon dioxide (CO_2), a favourable response, reducing methane by 0.61 g day^{-1} with no decline in MBW or CO_2, was reached from every methane trait once appropriate weights on MBW and CO_2 were included in the breeding goal, so the weighting rather than the trait definition determined the outcome.

Conclusions

Methane in sheep shows meaningful additive genetic variation across trait definitions. Ratio and residual methane traits remained strongly genetically correlated with absolute methane but differed in their genetic relationships with body size and growth, and therefore in the correlated responses expected in production traits. Methane can be reduced within a selection index without undesirable responses in other important traits, but the outcome is determined by how methane and these traits are weighted in the breeding goal rather than by the choice of methane trait definition.

Background

Reducing greenhouse gas emissions from ruminant livestock has become a priority for greenhouse gas mitigation strategies worldwide. Enteric methane emissions (hereafter referred to as methane emissions), produced during rumen fermentation, contribute approximately 30% of global anthropogenic methane emissions [1], making it one of the largest and most

challenging emission sources to address [2]. At the same time, rising global demand for animal-sourced protein also places increasing pressure on ruminant production systems to deliver sustained reductions in emissions without compromising productivity [3]. The trade-off between greenhouse gas mitigation and food production is particularly evident in grass-based systems, such as Ireland, where agriculture is the largest single sector source of national greenhouse gas emissions, with methane from enteric fermentation in cattle and sheep the dominant gas [4]. Concurrently, despite this high sectoral methane output, Ireland ranks among the most carbon-efficient regions globally for ruminant production when evaluated on a per-kilogram-of-product basis [5, 6]. This places particular emphasis on mitigation strategies capable of delivering sustained reductions in enteric methane emissions without reducing productivity or altering the fundamental structure of pasture-based systems.

Genetic selection for low-emitting animals has emerged as one of the most promising of these methane mitigation strategies, with moderate heritability reported for methane production [7] and the cumulative, long term nature of genetic gain [8, 9]. The recent development of high-throughput phenotyping technologies, most notably portable accumulation chambers (PACs), has enabled routine, large-scale collection of individual-animal methane phenotypes across both research and commercial flocks [10, 11]. These chambers allow groups of animals to be measured in approximately one hour under field conditions [12, 13]. Methane measurements from PACs are heritable (0.13–0.20 in grazing sheep) [10-12] and genetically correlated with respiration chamber (RC) methane output (rg = 0.62–1.00) [10, 14]. Portable accumulation chambers also record carbon dioxide (CO_2), a proxy for feed intake where intake cannot be measured directly [15]. Although each record is a short spot measurement with lower repeatability than RCs [10, 13], PACs rank animals accurately [16] and consistently across grazing and controlled environments [11], which is what genetic selection requires. Methane production has also been shown to have strong (positive) genetic and phenotypic correlations

with key production traits such as body weight, growth, and feed intake [7, 17-19]. As a result, direct selection for reduced absolute methane emissions risks inadvertently constraining animal performance.

The development of alternative methane trait definitions allows emissions to be quantified in a way that is less directly tied to an animal's body size and growth. Traits such as methane intensity, methane yield, and residual methane express emissions relative to an animal's metabolic demands or level of performance [18, 20]. These traits can identify animals that emit less methane than expected for their level of productivity and have therefore been proposed as a potential means of mitigating unfavourable genetic correlations between methane production and key production traits [21, 22]. Phenotypic analyses have shown that these alternative methane metrics can differ substantially in their relationships with production traits and with each other [23]. Both ratio and residual methane traits present known theoretical limitations, and, as noted by Lassen and Difford [24], the genetic parameters and responses to selection of these alternative definitions remain underexplored. The present study therefore estimates genetic parameters for alternative methane traits in sheep, including their relationships with each other and with key production traits, and explores how they respond to selection.

Methods

Gaseous emissions data

A total of 16,535 methane measurements were obtained from 8,354 sheep across 132 research and commercial flocks in Ireland between 2019 and 2025. Methane measurements were recorded using PACs following the protocol described by O'Connor et al. [25]. Animals were removed from feed at least one hour prior to measurement. Each animal was then randomly assigned to one of twelve PAC units. Once the chamber was sealed, gas concentrations were measured at entry and again after a 50-minute accumulation period using an RKI Eagle 2 gas monitor (Weatherall Equipment and Instruments Ltd., UK). Concentration changes for

methane (CH_4) and carbon dioxide (CO_2) were converted to daily output (g day^{-1}) using chamber-specific calibration equations.

Methane (g day^{-1}) and carbon dioxide production (g day^{-1}) measurements were first screened for outliers using a 1.5 × interquartile range criterion applied separately to both gases (n = 511 (3.09%) gaseous records removed). All gaseous records were assigned to a contemporary group of date of measurement, flock, and PAC measurement group (i.e., the same 12 animals measured at the same time-point). Only contemporary groups with at least 5 animals were retained, leaving 15,869 records from 8,185 animals.

Measurements were collected across multiple physiological stages, including lambs (both sexes, 63 to 365 days of age; 5,564 records from 3,034 animals), hoggets (nulliparous females aged 365 to 600 days; 1,812 records from 1,271 animals), and mature ewes (females with a recorded lambing event aged 431 days to 10.5 years; 7,828 records from 4,327 animals). A total of 25.40% of animals had multiple methane records both within or across physiological stages.

Additional phenotypic data

Additional phenotypic measurements were available for animals with methane records; these included live weight, average daily gain (ADG), muscle mass, and rumen volume, and were used to derive alternative methane traits.

Live weight

Live weight was recorded immediately prior to PAC measurement using a Prattley electronic weighing scale (O'Donovan Engineering Co. Ltd., Cork, Ireland) and was subjected to outlier removal (1.5 × interquartile range criterion). Metabolic body weight (live $weight^{0.75}$) was also derived for each animal. A total of 15,638 records from 8,002 animals were available for analysis.

Average daily gain

Average daily gain was derived for growing animals only (i.e. animals <660 days of age). All individual animal live weight records were obtained from the national sheep database operated by Sheep Ireland. For each methane measurement, live weights recorded within ±120 days of the PAC measurement date were considered. Average daily gain surrounding each methane measurement was estimated by fitting a linear regression of live weight on age in days, with the regression slope retained as the estimate of ADG (kg day^{-1}). A total of 4,316 ADG estimates were available from 2,658 growing animals after outlier removal (1.5 × interquartile range criterion).

Rumen volume and muscle mass

Individual animal computed tomography (CT) scan measurements taken within ±3 days of methane measurement were also available on a subset of animals. Computed tomography provides *in vivo* estimates of carcass tissue composition by classifying pixels in cross-sectional X-ray images according to their density, allowing fat, muscle and bone tissues to be quantified non-invasively as described by Clelland et al. [26]. From these scans, estimates of rumen volume (litres) and muscle mass (kg) were available. Following outlier screening (1.5 × interquartile range criterion), 766 CT measurements were available for both rumen volume (746 animals) and muscle mass (751 animals).

Additional data

Auxiliary data relating to animal age, breed composition, animal heterosis and recombination loss, and birth and rearing type were also available. The breed proportion of each animal for the six breeds with the greatest number of methane phenotypes (i.e., Belclare, Charolais, Cheviot, Suffolk, Lleyn, and Texel) was also calculated, together with the combined proportion of all other or unknown breeds. The general heterosis and recombination loss coefficients for each animal were also included [27, 28]:

$$het = 1 - \Sigma_{i=1}^{n} sire_i \times dam_i$$

$$rec = 1 - \Sigma_{i=1}^{n} (sire_i^2 + dam_i^2) / 2$$

where $sire_i$ and dam_i are the proportion of breed i in the sire and dam, respectively. For all growing animals the birth litter size (defined as whether the lamb was born as a single, twin, triplet or quadruplet), rearing litter size (defined as whether the lamb was reared as a single, twin or triplet), and dam parity (defined as 1, 2, 3, 4, 5, 6 or ≥7) were available. For ewes the birth and rearing type corresponding to the litter born to the ewe for the year of measurement were also available.

Methane trait definitions

A suite of methane and CO_2 traits was constructed (Table 1), comprising absolute gas traits (daily CH_4 and CO_2 production) and two groups of alternative methane traits: ratio-based traits and residual traits. In addition to daily (absolute) methane production (CH_4, g day^{-1}), a series of alternative methane traits were investigated, including ratio traits, where CH_4 was divided by a production or physiological trait, and residual traits, where CH_4 was regressed on one or more of these traits.

Absolute Gas Production

Methane production (CH_4) measured in g day^{-1} represented the total mass of enteric methane, released via eructation and flatulence, emitted per animal per day as captured by PAC measurement. Carbon dioxide production (CO_2), measured concurrently on the same PAC record and also expressed in g day^{-1}, was also analysed.

Ratio-Based Methane Traits

For all animals, two ratio traits were calculated: methane per kg of metabolic body weight, representing methane intensity (CH_4/MBW; g day^{-1} $kg^{-0.75}$), and the other representing methane as a proportion of total methane and carbon dioxide production (CH_4 ratio). The CH_4 ratio was calculated on a mass basis (i.e. both expressed in grams) as $CH_4/(CH_4+CO_2)$, with both gases expressed in g day^{-1}. For growing animals only, an additional three ratio traits were also computed: methane per kg of average daily gain (CH_4/ADG; g kg^{-1}), methane per kg

muscle mass (CH_4/MM; g day^{-1} kg^{-1}), and methane per litre rumen volume (CH_4/rumen; g day^{-1} L^{-1}).

Residual Methane Traits

Three residual methane traits (RMT) were derived: methane adjusted for metabolic body weight (RMT_{MBW}), methane adjusted for metabolic body weight and carbon dioxide production ($RMTMBW+_{CO_2}$), and methane adjusted for average daily gain (RMT_{ADG}; calculated in growing animals only). Residual methane traits were generated by regressing daily methane production (g day^{-1}) on the production trait of interest (metabolic body weight, carbon dioxide production; and where available, average daily gain). The residuals from each regression were extracted and used as the phenotype for the corresponding residual methane trait.

Genetic analysis

Variance components for all alternative methane traits were estimated using pedigree-based animal models fitted in ASReml version 4.2 [29]. Pedigree records were available for 36,449 animals (all phenotyped animals plus their recursively traced ancestors) and were used to define the additive genetic relationships among animals; 100% of phenotyped animals were represented in the pedigree, 88.5% had both parents known, the pedigree comprised 1,025 unique sires with a mean pedigree depth of 17.6 generations, and 73.1% of sires linked in at least two contemporary groups. All methane traits were analysed using the following linear mixed animal model:

$$y = \mu + sex + BR + CL + CV + LY + SU + TX + UN + het + rec + age + BTg + RTg + BT_e + RT_e + DP + CG + a + pe + e$$

where y was the phenotypic observation for the methane trait under analysis; μ was the overall mean; sex was the fixed class effect of sex (male or female); the fixed effects for breed proportion corresponding to Belclare (BR), Charolais (CL), Cheviot (CV), Lleyn (LY), Suffolk (SU), Texel (TX), and the combined proportion of all other or unknown breeds (UN), were

included as covariates; *het* was the heterosis coefficient; *rec* was the recombination loss coefficient; *age* was the age at measurement (weeks of age); *BTg* was the fixed class effect of birth litter size of the growing animal itself (single, twin, triplet or quadruplet); *RTg* was the fixed class effect of rearing litter size of the growing animal itself (single, twin or triplet); *BTe* was the fixed class effect of birth litter size recorded for the ewe in the year of measurement (single, twin, triplet or quadruplet); *RTe* was the fixed class effect of rearing litter size recorded for the ewe in the year of measurement (single, twin or triplet); *DP* was the fixed class effect of dam parity for growing animals; *CG* was the contemporary group fixed class effect of flock-date-lot of measurement; *a* was the random animal additive genetic effect; *pe* was the non-additive random animal permanent environmental effect; and *e* was the residual random error term, where $\boldsymbol{A}$ was the numerator relationship matrix, $\boldsymbol{I}$ was the identity matrix, σ^2_a was the direct additive genetic variance, σ^2_{pe} was the animal permanent environmental variance, and σ^2_e was the residual variance.

This model described above was used for both univariate and bivariate analyses. As animals of different ages and sizes were recorded, residual variance was expected to differ between contemporary groups, increasing with the group mean [30]. To account for heterogeneous residual variance, contemporary groups within each flock were ranked by mean CH_4 and split into three classes (low, medium and high methane emitting CGs), and a separate residual variance was fitted for each flock × class combination (41 classes; flocks with fewer than three contemporary groups contributed fewer classes). This heterogeneous residual model was used for CO_2 and for the five methane traits recorded across all physiological stages (CH_4, CH_4/MBW, CH_4 ratio, RMT_{MBW}, RMT_{MBW+CO_2}), and the heritability (h^2) and repeatability (t) were calculated for each class and averaged across classes weighted by the number of records. For the four traits restricted to a single physiological stage (CH_4/ADG, CH_4/MM, CH_4/rumen, RMT_{ADG}), a single homogeneous residual variance was used. For CH_4/MM, CH_4/rumen and

RMTADG, the permanent environmental effect was omitted from the model because of the small number of repeated records (the permanent environmental variance converged to zero). Bivariate analyses assumed a homogeneous residual variance.

Heritability (h^2) and repeatability (t) were calculated as:

$$h^2 = \sigma^2_a / (\sigma^2_a + \sigma^2_{pe} + \sigma^2_e)$$

$$t = (\sigma^2_a + \sigma^2_{pe}) / (\sigma^2_a + \sigma^2_{pe} + \sigma^2_e)$$

where σ^2_a, σ^2_{pe} and σ^2_e are as defined above; where the permanent environmental effect was omitted, t is not reported. Genetic and phenotypic correlations were obtained from the corresponding fitted (co)variance components, with standard errors derived by the delta method; computations were implemented in R version 4.5.3 [31].

Selection index analysis

To investigate the impact of methane traits on the response to selection, a selection index including CH_4, MBW and CO_2 was constructed.

Genetic parameters were taken from a trivariate animal model of CH_4, MBW and CO_2 fitted in ASReml [29] to the same 15,869 records, with the fixed effects of the univariate model, and with animal permanent environmental effects assumed uncorrelated between traits. Phenotypic variances (σ^2_p) are the sums of the additive, permanent environmental and residual components, and are therefore conditional on the fixed effects.

Index weights were derived using Smith–Hazel selection index theory [32, 33], with CH_4, MBW and CO_2 recorded on each selection candidate. The index weights are $\mathbf{b} = \mathbf{P}^{-1}\mathbf{Ga}$, where $\mathbf{P}$ and $\mathbf{G}$ are the phenotypic and additive genetic (co)variance matrices of the three traits and $\mathbf{a}$ contains the weight on each trait in the breeding goal. These goal weights are relative weights, not economic values: they define each methane trait as a combination of CH_4, MBW and CO_2, or are chosen to hold the MBW and CO_2 responses at zero, and only their ratios affect the index and responses. The predicted response in each trait is $\Delta G = i \cdot \boldsymbol{Gb}/\sqrt{(\boldsymbol{b}'\boldsymbol{Pb})}$, where i is the

selection intensity. Responses are reported per generation for a selection intensity (i) of 1. Responses for i = 1.7 (approximately the top 10% of candidates selected) and annual responses (assuming a generation interval of 2.8 years) were also calculated and reported in Supplementary Table S2.

Each methane trait was expressed as a breeding goal with weights on CH_4, MBW and CO_2. Absolute methane placed a weight on CH_4 only. The residual traits deducted MBW and CO_2 from CH_4 using either the genetic regression coefficients (genetic residual; 0.833 per $kg^{0.75}$ and -6.09×10^{-5} per g day^{-1}) or the phenotypic regression coefficients (phenotypic residual, corresponding to RMTMBW+CO_2; 0.423 per $kg^{0.75}$ and 0.00371 per g day^{-1}). The ratio traits CH_4/MBW and CH_4 ratio ($CH_4/(CH_4+CO_2)$) were linearised around the trait means using a first-order Taylor series approximation [34, 35], which is equivalent to deducting 0.810 × MBW and 0.0151 × CO_2 from CH_4, respectively. Linearised at the means, CH_4 ratio and CH_4/CO_2 give the same goal direction (0.0151 = mean CH_4 / mean CO_2), so the results for CH_4 ratio apply equally to CH_4/CO_2.

The responses attainable by any index of CH_4, MBW and CO_2 were first mapped (Figure 1), and a favourable region was defined as the set of responses in which CH_4 declined and neither MBW nor CO_2 declined. For each methane trait, the additional goal weights on MBW and CO_2 needed to reach a reference favourable response were then derived; the reference response was defined as the largest methane reduction attainable with MBW and CO_2 both held at zero response (Kempthorne and Nordskog [36]; related to the desired-gains approach [37]). For each methane trait, these weights and the resulting responses are given in Table 6.

Uncertainty in the index results was quantified by Monte Carlo simulation. The 15 (co)variance parameters of the trivariate model were resampled jointly from a multivariate normal distribution, using their REML estimates and the sampling (co)variance matrix from ASReml, so that both the standard errors of the estimates and the correlations between the traits were

accounted for. For each of 10,000 samples, **G** and **P** were rebuilt and the index results recomputed. Results are reported as point estimates with 95% intervals (2.5th and 97.5th percentiles of the samples).

The analysis assumes that genetic parameters are constant across generations, that selection is on the candidate's own records (no relatives' records or genomic information), and that all three traits are recorded on every candidate. Analyses were performed in R version 4.5.3 [31].

Results

Fixed-effect significance (Wald F-tests) and diagnostics of pedigree completeness and connectedness are presented in Supplementary Material S1. Texel and Suffolk proportions varied substantially within contemporary groups, whereas Cheviot and Lleyn proportions were largely confined to particular contemporary groups, so their breed effects cannot be completely separated from contemporary group effects. Most contemporary groups (82.1%) contained a single sex. The mean MBW was 22.10 (SD = 4.92) kg, mean live weight was 62.72 (SD = 18.18) kg, while mean ADG, muscle mass, and rumen volume were 0.18 (SD = 0.10) kg day^{-1}, 11.7 (SD = 3.24) kg, and 6.16 (SD = 1.49) L, respectively. Descriptive statistics for methane traits are presented in Table 1. Mean methane production was 17.9 (SD = 7.56) g day^{-1}, with methane intensity (CH_4/MBW) averaging 0.81 (SD = 0.29) g day^{-1} $kg^{-0.75}$. Mean carbon dioxide production was 1,189.6 (SD = 534.2) g day^{-1} (range 40.9–2,869.1). Methane expressed relative to average daily gain (CH_4/ADG) had a mean of 137.9 (SD = 189.3) g kg^{-1}, reflecting the small denominator in this ratio trait. The mean values for methane traits derived from CT measurements were 1.44 (SD = 0.48) g day^{-1} kg^{-1} for CH_4/MM and 2.69 (SD = 0.88) g day^{-1} L^{-1} for CH_4/rumen. Residual methane traits were centred around zero by construction, with phenotypic standard deviations of 6.53 g day^{-1} for RMTMBW, 5.32 g day^{-1} for RMTMBW+CO_2, and 4.75 g day^{-1} for RMTADG.

Table 1. Descriptive statistics for alternative methane trait definitions.

Trait group	Trait	No. records	No. animals	μ (SD)	Range
Absolute	CH_4	15,869	8,185	17.90 (7.56)	4.01 – 40.31
Absolute	CO_2	15,869	8,185	1189.64 (534.24)	40.92 – 2869.08
Ratio	CH_4/MBW	15,638	8,002	0.81 (0.29)	0.14 – 2.38
	CH_4 ratio	15,869	8,185	0.02 (0.01)	0.00 – 0.19
	CH_4/ADG	4,316	2,658	137.89 (189.27)	12.03 – 2058.33
	CH_4/MM	766	751	1.44 (0.48)	0.33 – 3.08
	CH_4/rumen	766	746	2.69 (0.88)	0.65 – 5.73
Residual	RMT_{MBW}	15,638	8,002	0.00 (6.53)	-20.48 – 21.33
	RMT_{MBW+CO_2}	15,638	8,002	0.00 (5.32)	-23.38 – 24.15
	RMT_{ADG}	4,316	2,658	0.00 (4.75)	-10.23 – 25.82

μ (SD) = mean (standard deviation). Trait abbreviations: CH_4 = daily methane output (g day^{-1}); CO_2 = daily carbon dioxide output (g day^{-1}); CH_4/MBW = methane per kg metabolic body weight (g day^{-1} $kg^{-0.75}$); CH_4 ratio = $CH_4/(CH_4+CO_2)$, mass basis; CH_4/ADG = methane per kg average daily gain (g kg^{-1}); CH_4/MM = methane per kg computed tomography (CT) muscle mass (g day^{-1} kg^{-1}); CH_4/rumen = methane per litre CT rumen volume (g day^{-1} L^{-1}); RMT_{MBW}, RMT_{MBW+CO_2} and RMT_{ADG} = residual methane (g day^{-1}) from regression of CH_4 on metabolic body weight; metabolic body weight and CO_2; and average daily gain, respectively.

Genetic parameters

Genetic parameters for the methane and CO_2 trait definitions are presented in Table 2. Genetic standard deviations ranged from 0.06 g day^{-1} $kg^{-0.75}$ for methane intensity (CH_4/MBW) to 35.92 g kg^{-1} for methane expressed relative to growth rate (CH_4/ADG), with corresponding genetic coefficient of variation (CVa) values of 7.44% and 26.05%, respectively. Methane production (CH_4) had a genetic standard deviation of 1.47 g day^{-1} and CVa of 8.20%, and carbon dioxide production (CO_2) had a genetic standard deviation of 117.18 g day^{-1} and CVa of 9.85%. The genetic standard deviation for methane expressed relative to muscle mass and rumen volume were 0.20 g day^{-1} kg^{-1} and 0.37 g day^{-1} L^{-1} with CVa values of 13.59% and 13.83%, respectively. Residual methane traits (RMT_{MBW}, RMT_{MBW+CO_2}, and RMT_{ADG}) had genetic standard deviations of 1.29, 1.41, and 1.55 g day^{-1}.

Heritability estimates ranged from 0.12 ± 0.03 for CH_4/ADG to 0.46 ± 0.11 for CH_4/MM (Table 2). Heritability estimates for CH_4 (0.19), CH_4/MBW (0.16), CH_4 ratio (0.19), RMT_{MBW} (0.16) and RMT_{MBW+CO_2} (0.17), from the heterogeneous-residual model, and for CH_4/ADG (0.12 ± 0.03) and RMT_{ADG} (0.28 ± 0.03) were broadly similar. The heritability of carbon dioxide production (CO_2), also estimated from the heterogeneous-residual model, was somewhat higher than that of most methane traits (0.25; repeatability 0.43). Methane expressed relative to muscle mass and rumen volume produced larger heritability estimates (0.46 ± 0.11 and 0.35 ± 0.11), although both were associated with larger standard errors, consistent with their much smaller sample sizes (766 CT records). Where the permanent environmental variance could be estimated, repeatability ranged from 0.20 (CH_4/MBW) to 0.43 (CO_2).

Table 2. Genetic parameters for methane and CO_2 trait definitions.

Trait group	Trait	σa	σpe	h² (SE)	t (SE)	CVa
Absolute	CH_4	1.47	0.61	0.19	0.22	8.20%
Absolute	CO_2	117.18	97.26	0.25	0.43	9.85%
Ratio	CH_4/MBW	0.060	0.034	0.16	0.20	7.44%
	CH_4 ratio	1.63×10^{-3}	1.39×10^{-3}	0.19	0.32	–
	CH_4/ADG	35.92 (4.73)	44.35 (3.68)	0.12 (0.03)	0.30 (0.02)	26.05%
	CH_4/MM	0.20 (0.03)	–	0.46 (0.11)	–	13.59%
	CH_4/rumen	0.37 (0.06)	–	0.35 (0.11)	–	13.83%
Residual	RMT_{MBW}	1.29	0.90	0.16	0.24	–
	RMT_{MBW+CO_2}	1.41	1.14	0.17	0.28	–
	RMT_{ADG}	1.55 (0.08)	–	0.28 (0.03)	–	–

σ_a = direct genetic standard deviation; σ_{pe} = animal permanent environmental standard deviation; h^2 = heritability; t = repeatability; CV_a = coefficient of genetic variation. Standard errors in parentheses. Trait abbreviations: CH_4 = daily methane output (g day^{-1}); CO_2 = daily carbon dioxide output (g day^{-1}); CH_4/MBW = methane per kg metabolic body weight (g day^{-1} $kg^{-0.75}$); CH_4 ratio = $CH_4/(CH_4+CO_2)$, mass basis; CH_4/ADG = methane per kg average daily gain (g kg^{-1}); CH_4/MM = methane per kg computed tomography (CT) muscle mass (g day^{-1} kg^{-1}); CH_4/rumen = methane per litre CT rumen volume (g day^{-1} L^{-1}); RMT_{MBW}, RMT_{MBW+CO_2} and RMT_{ADG} = residual methane (g day^{-1}) from regression of CH_4 on metabolic body weight; metabolic body weight and CO_2; and average daily gain, respectively. For CO_2, CH_4, CH_4/MBW, CH_4 ratio, RMT_{MBW} and RMT_{MBW+CO_2}, a heterogeneous

residual variance (41 classes) was modelled, whereas for CH_4/ADG, CH_4/MM, CH_4/rumen and RMTADG a single homogeneous residual variance was assumed. – = not estimated.

Correlations among methane traits

Genetic and phenotypic correlations between daily methane production (CH_4), CO_2, and each alternative methane trait definition are presented in Table 3. Phenotypically, daily methane was strongly correlated with the size-adjusted definitions (0.87 with both CH_4/MBW and CH_4/MM, 0.90 with RMT_{MBW}, 0.79 with RMT_{MBW+CO_2} and 0.71 with CH_4/rumen) and essentially perfectly correlated with RMT_{ADG} (1.00), but more weakly with CH_4/ADG (0.31) and CH_4 ratio (0.46).

Genetically, daily methane was strongly correlated with methane intensity (CH_4/MBW; 0.85 ± 0.02) and CH_4/MM (0.85 ± 0.04), moderately-to-strongly with CH_4/rumen (0.78 ± 0.10) and CH_4 ratio (0.71 ± 0.03), and only moderately with CH_4/ADG (0.38 ± 0.08). Notably, all three residual methane traits were strongly genetically correlated with absolute methane: RMT_{MBW} (0.86 ± 0.01), RMT_{MBW+CO_2} (0.78 ± 0.02) and RMT_{ADG} (1.00 ± 0.00), the last being genetically indistinguishable from daily methane. Daily methane was also moderately correlated with CO_2 production both phenotypically (0.35) and genetically (0.57 ± 0.04).

Table 3. Genetic and phenotypic correlations between daily methane production (CH_4), CO_2, and each alternative methane trait definition.

Trait	Genetic correlation with CH_4 (SE)	Phenotypic correlation with CH_4
CO_2	0.57 (0.04)	0.35
CH_4/MBW	0.85 (0.02)	0.87
CH_4 ratio	0.71 (0.03)	0.46
CH_4/ADG	0.38 (0.08)	0.31
CH_4/MM	0.85 (0.04)	0.87
CH_4/rumen	0.78 (0.10)	0.71
RMT_{MBW}	0.86 (0.01)	0.90
RMT_{MBW+CO_2}	0.78 (0.02)	0.79
RMT_{ADG}	1.00 (0.00)	1.00

Standard errors of genetic correlations in parentheses; standard errors of phenotypic correlations were all ≤0.02. Trait abbreviations: CH_4 = daily methane output (g day^{-1}); CO_2 = daily carbon dioxide output (g day^{-1}); CH_4/MBW = methane per kg metabolic body weight (g day^{-1} $kg^{-0.75}$); CH_4 ratio = $CH_4/(CH_4+CO_2)$, mass basis; CH_4/ADG = methane per kg average daily gain (g kg^{-1}); CH_4/MM = methane per kg computed tomography (CT) muscle mass (g day^{-1} kg^{-1}); CH_4/rumen = methane per litre CT rumen volume (g day^{-1} L^{-1}); RMT_{MBW}, RMT_{MBW+CO_2} and RMT_{ADG} = residual methane (g day^{-1}) from regression of CH_4 on metabolic body weight; metabolic body weight and CO_2; and average daily gain, respectively.

Correlations between methane and production traits

Phenotypic and genetic correlations between methane traits and production traits are presented in Tables 4 and 5, respectively. Phenotypically, CH_4 showed weak to moderate positive correlations with the production traits examined, including metabolic body weight (MBW; 0.36 ± 0.01), live weight (LW; 0.35 ± 0.01), ADG (0.17 ± 0.02), rumen volume (0.31 ± 0.04), and muscle mass (0.26 ± 0.04). As expected, several ratio-based methane traits showed negative phenotypic correlations with the denominator production traits, including methane intensity (CH_4/MBW) with MBW (−0.10 ± 0.01), CH_4/MM with muscle mass (−0.24 ± 0.04), CH_4/ADG with ADG itself (−0.25 ± 0.02), and CH_4/rumen with rumen volume (−0.45 ± 0.03). Residual methane traits showed contrasting patterns, with RMT_{MBW} negatively phenotypically correlated with MBW (−0.08 ± 0.01), whereas RMT_{ADG} showed a positive phenotypic correlation with its own adjustment trait, ADG (0.12 ± 0.02).

Table 4. Phenotypic correlations between methane traits and production traits (standard errors in parentheses).

Trait group	Trait	Metabolic body weight	Live weight	Average daily gain	Muscle mass	Rumen volume
Absolute	CH_4	0.36 (0.01)	0.35 (0.01)	0.17 (0.02)	0.26 (0.04)	0.31 (0.04)
Ratio	CH_4/MBW	-0.10 (0.01)	-0.10 (0.01)	-0.11 (0.02)	-0.04 (0.04)	0.13 (0.04)
	CH_4 ratio	-0.06 (0.01)	–	-0.01 (0.02)	-0.06 (0.04)	0.19 (0.04)
	CH_4/ADG	-0.00 (0.02)	-0.02 (0.02)	-0.25 (0.02)	0.26 (0.05)	0.20 (0.05)
	CH_4/MM	-0.09 (0.05)	-0.06 (0.05)	0.01 (0.05)	-0.24 (0.04)	0.13 (0.04)
	CH_4/rumen	–	0.04 (0.05)	0.00 (0.05)	-0.03 (0.04)	-0.45 (0.03)
Residual	RMT_{MBW}	-0.08 (0.01)	-0.16 (0.01)	0.02 (0.02)	-0.13 (0.05)	0.07 (0.05)

	RMT_{MBW+CO_2}	-0.02 (0.01)	-0.12 (0.01)	0.02 (0.02)	–	–
	RMT_{ADG}	0.17 (0.03)	0.47 (0.02)	0.12 (0.02)	0.21 (0.08)	0.21 (0.08)

Trait abbreviations: CH_4 = daily methane output (g day^{-1}); CO_2 = daily carbon dioxide output (g day^{-1}); CH_4/MBW = methane per kg metabolic body weight (g day^{-1} $kg^{-0.75}$); CH_4 ratio = $CH_4/(CH_4+CO_2)$, mass basis; CH_4/ADG = methane per kg average daily gain (g kg^{-1}); CH_4/MM = methane per kg computed tomography (CT) muscle mass (g day^{-1} kg^{-1}); CH_4/rumen = methane per litre CT rumen volume (g day^{-1} L^{-1}); RMT_{MBW}, RMT_{MBW+CO_2} and RMT_{ADG} = residual methane (g day^{-1}) from regression of CH_4 on metabolic body weight; metabolic body weight and CO_2; and average daily gain, respectively. Muscle mass and rumen volume were measured by CT. Standard errors in parentheses; – = not estimated.

Genetically, CH_4 showed a moderate-to-strong positive genetic correlation with metabolic body weight (MBW; 0.72 ± 0.03), live weight (0.71 ± 0.03), ADG (0.51 ± 0.07), muscle mass (0.42 ± 0.12), and rumen volume (0.68 ± 0.15). Ratio-based methane traits were generally negatively genetically correlated with the denominator production traits, including methane intensity (CH_4/MBW) with MBW (−0.27 ± 0.04), CH_4/ADG with ADG itself (−0.81 ± 0.08), and CH_4/rumen with rumen volume (−0.14 ± 0.30, a correlation with a wide confidence interval). Methane ratio showed weak to moderate negative genetic correlations with MBW (−0.21 ± 0.04), ADG (−0.04 ± 0.06), and muscle mass (−0.23 ± 0.19). Among the residual methane traits, RMT_{MBW}'s genetic correlation with MBW was −0.28 ± 0.04, and RMT_{MBW+CO_2}'s with MBW was −0.11 ± 0.04, whereas RMT_{ADG} showed a moderate positive genetic correlation with its own adjustment trait, ADG (0.42 ± 0.07).

Table 5. Genetic correlations between methane traits and production traits (standard errors in parentheses).

Trait group	Trait	Metabolic body weight	Live weight	Average daily gain	Muscle mass	Rumen volume
Absolute	CH_4	0.72 (0.03)	0.71 (0.03)	0.51 (0.07)	0.42 (0.12)	0.68 (0.15)
Ratio	CH_4/MBW	-0.27 (0.04)	-0.27 (0.03)	-0.31 (0.08)	-0.10 (0.16)	-0.31 (0.22)
	CH_4 ratio	-0.21 (0.04)	–	-0.04 (0.06)	-0.23 (0.19)	0.13 (0.30)
	CH_4/ADG	-0.12 (0.07)	-0.19 (0.08)	-0.81 (0.08)	-0.02 (0.21)	0.06 (0.22)
	CH_4/MM	-0.26 (0.14)	-0.21 (0.12)	-0.04 (0.13)	–	0.22 (0.30)

	CH_4/rumen	–	0.12 (0.12)	-0.04 (0.12)	0.16 (0.22)	-0.14 (0.30)
Residual	RMT_{MBW}	-0.28 (0.04)	-0.44 (0.06)	0.01 (0.05)	-0.52 (0.48)	0.39 (0.42)
	RMT_{MBW+CO_2}	-0.11 (0.04)	-0.36 (0.05)	0.02 (0.05)	–	–
	RMT_{ADG}	0.31 (0.11)	0.94 (0.07)	0.42 (0.07)	0.37 (0.28)	0.37 (0.28)

Trait abbreviations: CH_4 = daily methane output (g day^{-1}); CO_2 = daily carbon dioxide output (g day^{-1}); CH_4/MBW = methane per kg metabolic body weight (g day^{-1} kg$^{-0.75}$); CH_4 ratio = $CH_4/(CH_4+CO_2)$, mass basis; CH_4/ADG = methane per kg average daily gain (g kg^{-1}); CH_4/MM = methane per kg computed tomography (CT) muscle mass (g day^{-1} kg^{-1}); CH_4/rumen = methane per litre CT rumen volume (g day^{-1} L^{-1}); RMT_{MBW}, RMT_{MBW+CO_2} and RMT_{ADG} = residual methane (g day^{-1}) from regression of CH_4 on metabolic body weight; metabolic body weight and CO_2; and average daily gain, respectively. Muscle mass and rumen volume were measured by CT. Standard errors in parentheses; – = not estimated.

Selection index analysis

Genetic parameters of CH_4, MBW and CO_2

Genetic parameters from the trivariate analysis of CH_4, MBW and CO_2, used for all index calculations, are given in Supplementary Table S3.

The favourable region

Figure 1 shows every response attainable by an index of CH_4, MBW and CO_2. The favourable region within it, in which CH_4 is reduced without a decline in MBW or CO_2, exists, and along its boundary the correlated CO_2 response was positive (+10.1 to +64.5 g day^{-1}). The region is small, covering about 6.5% (95% interval 4.9–7.8%) of the area of attainable responses in Figure 1, because CH_4 is strongly genetically correlated with both MBW and CO_2. When a zero response was assumed for both MBW and CO_2, methane was reduced by 0.61 g day^{-1} (95% interval 0.46–0.75), 46% (35–54%) of the largest attainable methane reduction (−1.34 g day^{-1}; −1.46 to −1.22). Holding MBW alone at zero gave a similar reduction (0.63 g day^{-1}) but increased CO_2 by 10.0 g day^{-1}. Within the region, the weighting sets the balance between the three traits: an index that increased MBW by 0.12 kg$^{0.75}$ and CO_2 by 20.7 g day^{-1} still reduced CH_4 by 0.52 g day^{-1}, and some methane reduction remained possible up to an MBW increase of about 0.65 kg$^{0.75}$ (0.52–0.77), or a CO_2 increase of about 75 g day^{-1} (63–86) (Figure 1).

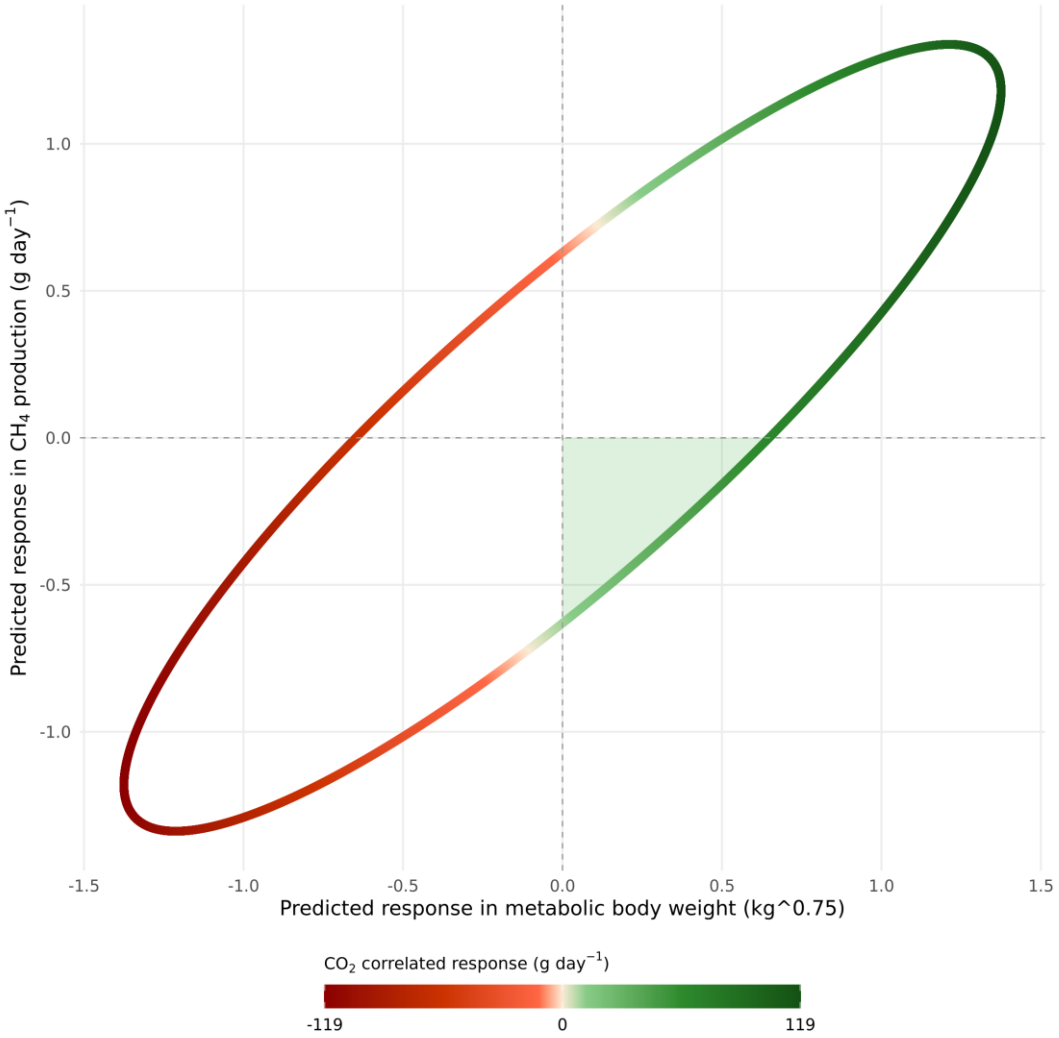


Figure 1. Responses in methane (CH_4) and metabolic body weight (MBW) attainable per generation (selection intensity 1) by any index of CH_4, MBW and CO_2. Colour shows the correlated response in CO_2 (g day^{-1}). The shaded area is the favourable region, in which CH_4 is reduced and neither MBW nor CO_2 is reduced, shown in the CH_4–MBW plane; the CO_2 response along its boundary is positive (+10.1 to +64.5 g day^{-1}). The reference favourable index (the largest CH_4 reduction with MBW and CO_2 held at zero response) lies on the region's MBW = 0 edge at a CH_4 response of −0.61 g day^{-1}.

The largest methane reduction attainable without any change in MBW or CO_2 (−0.61 g day^{-1} CH_4 per generation) could be reached from a breeding goal built on any of the methane traits, provided appropriate additional weights were placed on MBW and CO_2 (Table 6). All five

methane traits then led to the same index and the same responses. The outcome was therefore determined by the weights on MBW and CO_2, not by which methane trait the goal was built on.

Table 6. Additional breeding-goal weights on MBW and CO_2 needed to reach the reference favourable response (CH_4 reduced, MBW and CO_2 held at zero response) when each methane trait forms the breeding goal.

Methane term in the breeding goal	Additional weight on MBW (per $kg^{0.75}$)	Additional weight on CO_2 (per g day^{-1})	Response: ΔCH_4 (g day^{-1}) / ΔMBW ($kg^{0.75}$) / ΔCO_2 (g day^{-1})
CH_4	+1.21	−0.0040	−0.61 / 0 / 0
CH_4/MBW (linearised)	+0.40	−0.0040	−0.61 / 0 / 0
CH_4 ratio (linearised)	+1.21	−0.0191	−0.61 / 0 / 0
Genetic residual	+0.37	−0.0040	−0.61 / 0 / 0
Phenotypic residual (RMT_{MBW+CO_2})	+0.78	−0.0078	−0.61 / 0 / 0

Weights are per −1 on the methane term, i.e. the goal is −(methane term) + w(MBW)·MBW + w(CO_2)·CO_2. They are relative breeding-goal weights (a), not economic values: only their ratios matter, and all five goals give the same index and the same responses. They differ only because each methane term already carries some weight on MBW and CO_2 (ratios linearised around the trait means; residual regression coefficients as in Methods). The corresponding selection index weights (b) are given in Supplementary Table S4. Abbreviations: CH_4 = daily methane output (g day^{-1}); MBW = metabolic body weight ($kg^{0.75}$); CO_2 = daily carbon dioxide output (g day^{-1}); CH_4 ratio = $CH_4/(CH_4+CO_2)$, mass basis; RMT_{MBW+CO_2} = residual methane from regression of CH_4 on MBW and CO_2. Responses are per generation at selection intensity 1.

Discussion

The present study estimated genetic parameters for a comprehensive set of methane trait definitions in pasture-based sheep, comparing absolute, ratio-based, and residual methane traits. Multiple methane traits are used across the literature to quantify emissions, but their statistical properties, and their genetic relationships with production traits, are not always made explicit. Lassen and Difford [24] highlighted the need for further empirical work on the genetic parameters of ratio- and residual-based methane traits. This study addresses that gap and extends it by examining how these traits respond to selection within a selection index.

Genetic parameters of methane trait definitions

Heritability estimates across the nine methane trait definitions ranged from 0.12 to 0.46, indicating that methane production carries meaningful additive genetic variation whichever way it is expressed. These estimates lie within the range reported for methane traits in sheep [10, 38], beef [7, 19] and dairy cattle [9, 39], and are comparable in magnitude to traits already used in livestock breeding programmes, such as somatic cell count [40], which has delivered sustained genetic gain in dairy cattle once incorporated into a breeding objective [41]. Feed intake is the main determinant of an animal's daily methane output [42, 43]. Where intake is recorded, methane is therefore commonly expressed per unit of intake (methane yield) or residualised on intake [22, 44]. In grazing sheep, however, feed intake is rarely measured at scale, and scaling or residualising methane against body size, growth or production traits serves as a practical proxy for that missing feed-intake information [15, 18, 20]. The adjustment traits used here (MBW, ADG, muscle mass, rumen volume and CO_2) are also biologically relevant in their own right. Maintenance energy requirements, and hence much of an animal's feed intake, are proportional to metabolic body weight, so MBW captures the size-driven component of intake. Average daily gain describes the rate of growth and muscle mass the product of growth, so methane expressed relative to ADG reflects emissions relative to growth rate, and methane relative to muscle mass approximates emissions per unit of saleable product. Rumen volume reflects fermentative capacity and retention time [45]. Carbon dioxide output reflects total metabolic activity and is closely related to feed intake in PAC-measured sheep [15], therefore CH_4 ratio and RMT_{MBW+CO_2} approximate methane relative to intake.

In sheep, the heritability of daily methane was similar to those previously estimated using PAC in lambs and ewes (0.19–0.20) [10], but lower than respiration-chamber estimates [38]. In cattle, daily methane heritability measures range from 0.15 to 0.42 in beef [7, 19] and from 0.16 to 0.33 in dairy [9, 39], with the highest beef estimates from GreenFeed measurements [7, 19]. This is consistent with a single short PAC measurement capturing less of an animal's

480 average emission than repeated measurements [13, 25]. The heritability of CH_4 ratio fell within
481 the range of 0.17–0.25 reported by Jonker et al. [10]. Methane relative to product (0.21–0.66)
482 and residual methane (0.38–0.46) were generally more heritable in cattle [9, 19] than the
483 corresponding traits presented here, which adjust for body size, growth and CO_2 as proxies for
484 intake rather than for measured intake or output.

485 ## Genetic relationships with body size, growth and CO_2

486 Genetically, the alternative methane definitions were strongly correlated with absolute methane
487 in most cases (rg = 0.71–1.00), indicating that they share much of the same underlying additive
488 genetic variation rather than capturing a materially distinct trait. Methane per kg of average
489 daily gain (CH_4/ADG) was the most genetically distinct definition (rg = 0.38 with absolute
490 methane), consistent with the ratio being dominated by its small and highly variable
491 denominator, ADG (rg = −0.81), which is also estimated over a window of up to ±120 days
492 rather than at the time of the methane measurement. Daily methane was also strongly and
493 positively genetically correlated with body size and growth: metabolic body weight (rg = 0.72),
494 live weight (rg = 0.71), and average daily gain (rg = 0.51). Ratio and residual methane traits
495 did not remove these genetic relationships. Methane intensity and the residual methane traits
496 were negatively genetically correlated with their own denominator or adjustment trait, while
497 remaining strongly genetically correlated with absolute methane itself (rg = 0.78–1.00 for the
498 three residual traits). This is consistent with the general statistical behaviour of ratio and
499 residual variables described by Kennedy et al. [46] and Kronmal [47]. This is expected, as a
500 phenotypic residual is not genetically independent of the traits it is regressed on unless the
501 regression is applied at the genetic level.
502 Carbon dioxide production was moderately heritable (h^2 = 0.25) and strongly genetically
503 correlated with both daily methane (rg = 0.57 ± 0.04) and metabolic body weight (rg = 0.83 ±
504 0.02). Carbon dioxide output reflects total heat production and hence energy intake, and CO_2

is strongly correlated with feed intake in PAC-measured sheep [15, 48] and in cattle [49, 50]. However, more efficient animals produce less CO_2 per unit of intake [51], which limits its accuracy as a proxy for intake.

Heterogeneous residual variance

Genetic evaluations can assume a common residual variance across contemporary groups. This assumption is rarely met, and when it is violated, records from more variable groups receive more weighting and variance components and breeding values are biased [30, 52, 53]. It is particularly relevant to PAC methane: between-day and between-group variation in PAC measurements is large [25], emissions differ with site, pasture and protocol [12, 13], and the present data combined animals of different ages and physiological stages. Accounting for this heterogeneity lowered the heritability of methane, CH_4/MBW, RMT_{MBW} and RMT_{MBW+CO_2}, and of CO_2, relative to a single pooled residual variance. However, classifying the residual variance as either homogeneous or heterogeneous had little impact on the genetic correlations, for example the genetic correlation between methane and methane intensity was 0.85 in the full data, and also 0.85 in both young and mature animals when analysed separately, and the young-versus-mature genetic correlation for methane was 0.99. Residual heterogeneity therefore mainly affects heritability estimates and should be accounted for whenever genetic parameters are estimated from PAC methane recorded across physiological stages. However, only 25.4% of animals had more than one methane record, which limits the precision of the permanent environmental variance and prevented its estimation for the computed-tomography traits and RMT_{ADG}.

Methane traits in a selection index

Alternative methane traits have been frequently proposed as selection criteria [22, 44, 54, 55], but the central result of this index analysis is that the choice of methane trait is of secondary importance. A favourable region, in which methane is reduced without a decline in body size

or CO_2, does exist, and every methane trait examined, whether absolute, ratio or residual, reached the same response within it once appropriate weights on MBW and CO_2 were included in the breeding goal; all then led to the same index (Table 6). A ratio or residual trait is, in effect, a fixed weighting of methane against its denominator or adjustment trait [34, 46]. It is therefore the weighting of methane, body size and CO_2 in the breeding goal, not the methane trait definition, that determines whether methane can be reduced without reducing size.

This agrees with the observation that direct multi-trait selection can match or outperform selection on a ratio [56-58], and with the long-standing debate over ratio, residual and index selection for feed efficiency [59, 60]. Within the region, the balance can be shifted towards slightly higher body size or intake, as proxied by CO_2, at the cost of a smaller methane reduction; the reference index, which holds both at zero response, is the neutral choice. The weights reported here depend on the genetic parameters and are specific to this population. They were also treated as constant across generations, although genetic parameters change under selection [61]. In the present study, responses were calculated per generation at a fixed selection intensity, whereas realised selection intensity depends on the breeding programme, the recording of methane on only part of the population, and the use of genomic information, none of which were modelled.

The weights on MBW and CO_2 in the present index are not economic values but those required to reduce methane without changing MBW or CO_2, similar to a desired-gains approach [37]. An economic breeding goal would instead weight each trait, including methane, by its economic value [62-64], but the same principles apply. Economic values on an alternative methane trait are subject to the same constraints identified in this analysis: the trait already carries implicit weights on its component traits (here MBW and CO_2), which must be accounted for when those traits are weighted, and the correlated responses of the index should be checked against the attainable responses (Figure 1) to avoid an unintended cost to production.

Conclusions

This study estimated genetic parameters for nine alternative methane trait definitions in a large, pasture-based sheep population, and shows that these definitions differ meaningfully in their genetic parameters and in their genetic relationships with body size and growth. All definitions were heritable, with estimates ranging from low to moderate (h^2 = 0.12–0.46). Ratio and residual methane traits remained strongly genetically correlated with absolute daily methane, showing that they share much of the same underlying genetic variation rather than representing a genetically distinct trait. Daily methane itself was strongly and positively genetically correlated with body size and growth, and scaling or residualising methane redistributed this genetic association rather than removing it. When a methane trait is used on its own, the choice of definition therefore determines the direction and magnitude of the genetic relationship between methane and production.

The selection index analysis shows, however, that this choice matters far less than how methane is weighted against body size and CO_2 or indeed any other trait of importance. In an index of methane, metabolic body weight and CO_2, once appropriate weights on body size and CO_2 were included, every trait, whether absolute, ratio or residual, reached the same favourable response and the same index. The weighting of methane, body size and CO_2 in the breeding goal, not the methane trait definition, therefore determines whether methane can be reduced without constraining performance. In practice these weights would come from economic values, and the same framework identifies the minimum value that must be placed on methane, relative to production traits, for an economic index to achieve this.

Declarations

Ethics approval and consent to participate

Data were generated on growing animals and ewes from 132 Irish sheep flocks. Data collection was approved by the Teagasc Animal Ethics Committee (TAEC2020-252; TAEC2020-258;

TAEC0323-374) and the Health Protection Regulatory Authority (AE19132/P112; AE19132/P114; AE19132/P181).

Consent for publication

Not applicable.

Availability of data and materials

The datasets analysed in the current study are not publicly available as they contain commercially sensitive information collected from farmers and private enterprises. Supplementary materials, including full variance components and complete correlation matrices, are available via Zenodo [DOI to be added upon acceptance].

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

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Supplementary Material

S1. Fixed-effect significance, pedigree completeness and connectedness.

Table S1a. Significance of fixed effects (Wald F-tests) in the univariate models of the component traits.

Fixed effect	CH_4	CO_2	MBW	ADG	Muscle mass	Rumen volume	Live weight

sex	ns	ns	***	***	***	*	***
BR	ns	ns	ns	ns	***	***	ns
CL	***	ns	***	**	***	***	***
CV	ns	ns	**	ns	ns	ns	*
LY	ns	ns	ns	ns	ns	ns	ns
SU	*	ns	***	ns	***	***	***
TX	*	***	***	ns	***	ns	***
Other or unknown breeds	***	***	***	***	***	***	***
het	*	ns	ns	***	***	ns	ns
rec	ns	ns	***	**	***	ns	***
age	*	**	***	***	***	***	***
BTg	***	ns	***	*	ns	ns	***
RTg	***	***	***	ns	***	ns	***
BTe	***	***	***	–	–	–	***
RTe	**	ns	***	–	–	–	***
DP	ns	ns	*	ns	ns	ns	**

* P < 0.05; ** P < 0.01; *** P < 0.001; ns, not significant (P ≥ 0.05); –, term not in the model for that trait. Fixed effects as defined in the model equation (Methods); other or unknown breeds is the proportion of breeds other than the six named, fitted as a covariate.

Table S1b. Pedigree completeness and connectedness of the phenotyped animals.

Statistic	Value
Phenotyped animals represented in the pedigree	8,185 (100%)
Both parents known (%)	88.5
Neither parent known (%)	5.3
Traced pedigree depth, generations: mean (median, maximum)	17.6 (19, 34)
Unique sires	1,025
Unique dams	5,216
Sires with progeny in ≥ 2 contemporary groups (%)	73.1
Sires with progeny in ≥ 5 contemporary groups (%)	43.8

Contemporary groups linked by the most widely used sire	205

606

607 **S2.** Predicted responses at selection intensity 1.7, per generation and per year (generation
608 interval 2.8 years). Responses of the reference favourable index (Table 6) multiplied by 1.7
609 and by 1.7/2.8 = 0.607, respectively.

Index	ΔCH₄ (g day⁻¹) per generation	ΔMBW per generation	ΔCO₂ (g day⁻¹) per generation	ΔCH₄ per year	ΔMBW per year	ΔCO₂ per year
Reference: CH₄ reduced, MBW and CO₂ held at zero	−1.04	0.00	0.0	−0.37	0.00	0.0

610

611 **S3.** Genetic parameters from the trivariate analysis of CH₄, MBW and CO₂ used in the selection
612 index analysis.

Trait	Mean	σa	σp	h² (SE)	rg with CH₄ (SE)	rg with MBW (SE)	rg with CO₂ (SE)
CH₄ (g day⁻¹)	17.90	2.10	4.02	0.273 (0.017)	–	0.69 (0.03)	0.57 (0.03)
MBW ($kg^{0.75}$)	22.10	1.75	2.30	0.582 (0.019)	0.69 (0.03)	–	0.83 (0.02)
CO₂ (g day⁻¹)	1189.64	169.8	265.6	0.409 (0.017)	0.57 (0.03)	0.83 (0.02)	–

613 σa and σp are genetic and phenotypic standard deviations from the fitted variance components; σp is conditional
614 on the fixed effects. Phenotypic correlations: CH₄–MBW 0.36, CH₄–CO₂ 0.37, MBW–CO₂ 0.50 (SE ≤ 0.01). h²
615 = heritability.
616

S4. Selection index weights (b, applied to phenotypes; relative weights, not economic values) giving the reference favourable response (CH_4 reduced, MBW and CO_2 held at zero response), written for each methane trait.

Goal coordinates	Weight on methane term (per g day^{-1})	Weight on MBW (per kg$^{0.75}$)	Weight on CO_2 (per g day^{-1})
Absolute CH_4 goal (CH_4, MBW, CO_2 as themselves)	−0.1618	0.1348	-9.856×10^{-6}
Genetic-residual goal (weights on residual, MBW, CO_2)	−0.1618	0	0
Phenotypic-residual goal (weights on residual, MBW, CO_2)	−0.1618	0.0664	-6.095×10^{-4}

The methane term is CH_4, the genetic residual or the phenotypic residual, respectively; all three rows are the same index and give the same responses (Table 6). Weights are selection index weights (b) on the phenotypes, not economic values; multiplying all three by a constant gives the same index and responses. Expressed as relative emphasis (index weight × phenotypic standard deviation from the trivariate model, as a percentage of the summed absolute values), the index places −68% on CH_4, +32% on MBW and −0.3% on CO_2, identically for all three rows. In each row, the MBW and CO_2 weights are those added on top of any already built into the methane term, which is why they are zero for the genetic residual. Weights are on unscaled phenotypes; the CO_2 weight is numerically small because CO_2 is measured in g day^{-1}.

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